

GRADE 9 MATHEMATICS · ALBERTA CURRICULUM · MEASUREMENT & GEOMETRY**Circle Geometry**

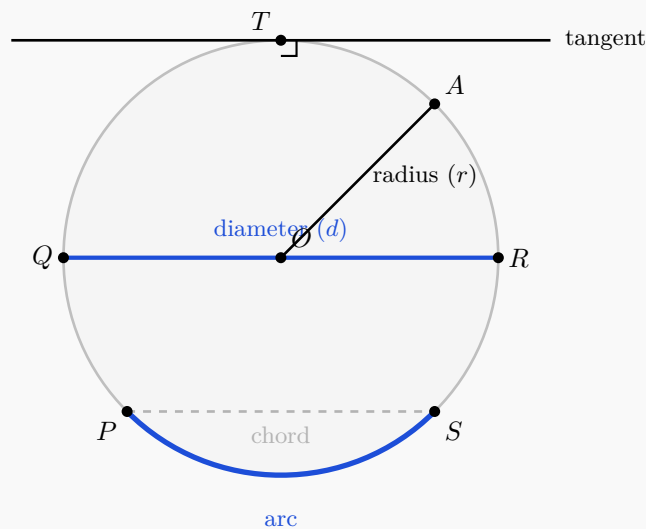
Guided Notes Package

Name: _____

Date started: _____

Pre-Knowledge Check*No marks. No correction. Be honest — write exactly what you think right now.*

Label any parts of a circle that you already know on this diagram:



What do you think an “inscribed angle” is? Make your best guess.

What angle do you think is formed when a radius meets the edge of a circle?

You will come back to this page at the end of the unit. Do not erase your answers.

Section 1 — Circle Vocabulary

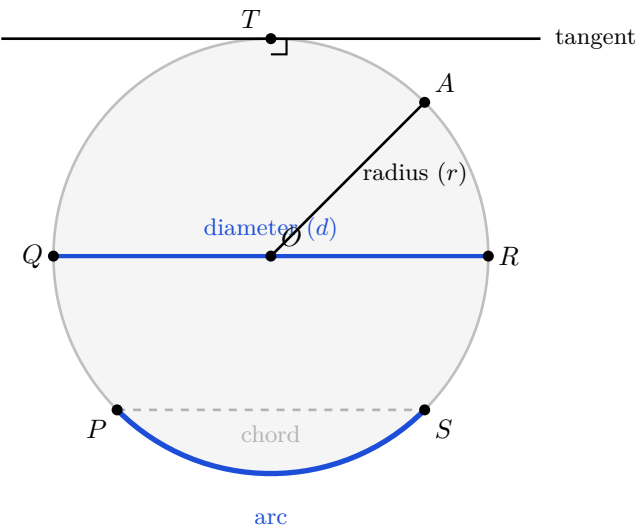
Learning target: I can identify and name all major parts of a circle and use precise geometric language when describing circle relationships.

VOCABULARY

Term	Definition in your own words	One example / drawing
Centre		
Radius		
Diameter		
Chord		
Arc		
Central angle		
Inscribed angle		
Tangent		
Point of tangency		
Semicircle		

GUIDED NOTES

Label each part on the diagram below:



Fill in the labels: P, Q, O, R, T, and the two unlabelled lines.

Fill in the blanks:

The **centre** of a circle is the point _____ from all points on the circle.

The **radius** (r) goes from the _____ to any point on the circle.

The **diameter** (d) is the _____ chord — it passes through the _____.

$$d = ___ \times r \qquad r = d \div ___$$

A **chord** is a line segment with both endpoints on the _____. (*The diameter is a special chord.*)

An **arc** is _____ of the circle between two points.

Wrong idea: “Any chord that goes through the circle is a diameter.”

Why it fails: A diameter is a chord that passes specifically through the *centre*. Most chords do not pass through the centre and are shorter than the diameter. Every diameter is a chord, but not every chord is a diameter.

In the diagram above, QR passes through the centre O, so QR is a _____.

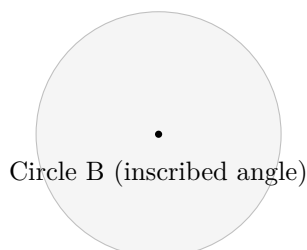
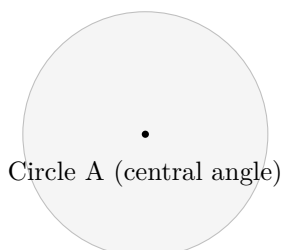
A chord that does not pass through O would be _____ the diameter.

Central angle vs. inscribed angle:

A **central angle** has its vertex at the _____ of the circle.

An **inscribed angle** has its vertex on the _____ of the circle, with both sides being chords.

Draw an example of each:



The **arc intercepted by an angle** is the arc that lies _____ the angle.

Tangent:

A **tangent** is a line that touches the circle at exactly _____ point (the point of tangency).

At the point of tangency, the tangent is _____ to the radius. (*We will prove this in Section 4.*)

Is a tangent the same as a chord? _____ Explain: _____

Naming arcs:

A **minor arc** is the _____ arc between two points (less than a semicircle).

A **major arc** is the _____ arc (more than a semicircle).

A **semicircle** is exactly _____ of the circle — it is the arc cut by a _____.

Retrieval Check — *Close your notes*

1. What is the difference between a chord and a diameter?

2. What is the difference between a central angle and an inscribed angle?

3. In a circle with diameter 14 cm, what is the radius?

4. A tangent line touches a circle at one point. What property does it have relative to the radius at that point?

5. (*Stretch*) Can two different chords in the same circle be equal in length? What would this tell you about their distances from the centre?

Self-check: ☐ I could teach it ☐ Getting there ☐ Not yet

Section 2 — Chord Properties

Learning target: I can apply the two chord properties — the perpendicular bisector property and the equal chords/equal distance property — to solve problems.

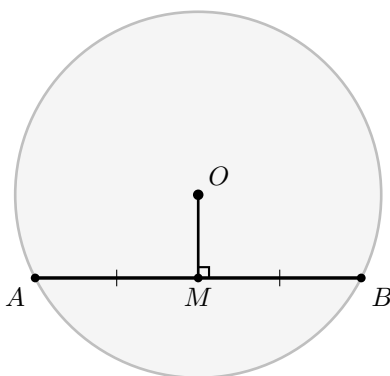
VOCABULARY

Term	Definition in your own words	One example
Perpendicular bisector		
Bisect		
Equidistant		
Distance from centre to chord		

GUIDED NOTES

Property 1: The Perpendicular from the Centre Bisects the Chord

If a line from the _____ of a circle is perpendicular to a chord, then it _____ that chord.



In symbols: If $OM \perp AB$, then $AM =$ _____

The converse is also true: If a line from the centre bisects a chord, then it is _____ to the chord.

Worked Example 1

A chord AB is 16 cm long. The perpendicular from centre O meets AB at M . Find AM .

Since $OM \perp AB$ and passes through the centre, it bisects AB .

$$AM = \frac{16}{2} = 8 \text{ cm}$$

Now extend — use the Pythagorean theorem. If the radius is 10 cm, find OM .

Right triangle OMA : $OA =$ _____ (radius), $AM = 8$, $OM = ?$

$$OA^2 = OM^2 + AM^2$$

$$10^2 = OM^2 + 8^2$$

$$OM^2 = 10^2 - 8^2 = 36$$

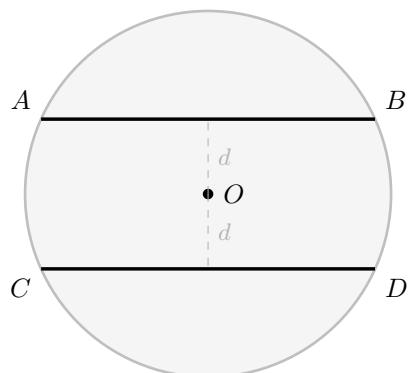
$$OM = 6 \text{ cm}$$

Explain: Which side is the hypotenuse in triangle OMA? How do you know?

Property 2: Equal Chords Are Equidistant from the Centre

In the same circle:

- If two chords are _____ in length, then they are _____ from the centre.
- Conversely: if two chords are the same distance from the centre, they are _____.



If $AB = CD$, then $d_1 =$ _____

Wrong idea: “The longest chord should be farthest from the centre.”

Why it fails: It’s the opposite. The chord *closest* to the centre is the longest. The diameter — the longest chord of all — passes through the centre at distance 0. As a chord moves farther from the centre, it gets shorter and shorter until it disappears at the edge.

Worked Example 2

Two chords PQ and RS are each 24 cm long in a circle with radius 13 cm.

Since $PQ = RS$, they are equidistant from the centre.

Let M be the midpoint of PQ. $PM = \text{_____}/2 = \text{_____}$

Right triangle OMP: $OP = 13$ (radius), $PM = 12$, $OM = ?$

$$13^2 = OM^2 + 12^2 \implies OM = \text{_____}$$

Therefore both chords are _____ cm from the centre.

Retrieval Check — Close your notes

1. State the perpendicular bisector chord property in your own words.
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2. A chord is 18 cm long. The perpendicular from the centre has length 12 cm. Find the radius. (Draw a diagram first.)
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3. Two chords in the same circle are both 10 cm long. What can you conclude about their distances from the centre?

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4. In a circle with radius 5 cm, a chord is 3 cm from the centre. How long is the chord?

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5. (*Stretch*) A chord is 8 cm from the centre of a circle with radius 17 cm. Another chord is 15 cm from the centre. Which chord is longer? By how much?

Self-check: ☐ I could teach it ☐ Getting there ☐ Not yet

Connection: The perpendicular bisector of a chord passes through the centre. You can use this to find the centre of any circle if you don't know where it is. How would you do that if given only the circle drawn on paper? Describe the steps.

Section 3 — Central Angle and Inscribed Angle Theorem

Learning target: I can apply the inscribed angle theorem to find unknown angles, and explain why inscribed angles subtending the same arc are equal.

VOCABULARY

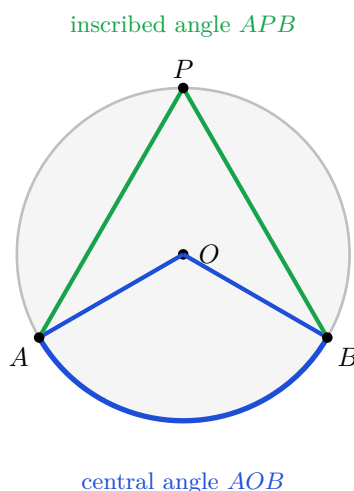
Term	Definition in your own words	One example
Inscribed angle theorem		
Subtend		
Same arc		

GUIDED NOTES

The Inscribed Angle Theorem:

The inscribed angle is _____ the central angle that subtends the same arc.

$$\text{Inscribed angle} = \frac{1}{2} \times \text{Central angle} \quad \text{OR} \quad \text{Central angle} = \underline{\hspace{1cm}} \times \text{Inscribed angle}$$



Angle AOB (central) = _____ $^\circ$ Angle APB (inscribed, same arc AB) = _____ $^\circ$

Worked Example 1

Central angle $AOB = 80^\circ$. Find inscribed angle APB subtending the same arc.

$$\text{Inscribed angle} = 80^\circ \div \underline{\hspace{1cm}} = \underline{\hspace{1cm}}^\circ$$

Worked Example 2

Inscribed angle $APB = 35^\circ$. Find central angle AOB subtending the same arc.

$$\text{Central angle} = 2 \times 35^\circ = \underline{\hspace{1cm}}^\circ$$

Explain: Why does being at the centre make the central angle “see” the arc at twice the angle?

Corollary: Inscribed angles subtending the same arc are equal.

$APB = AQB = \frac{1}{2} \times AOB$ regardless of where P and Q are on the major arc.

Wrong idea: “Any angle drawn inside a circle is 90° .”

Why it fails: Only inscribed angles that subtend a *diameter* equal 90° . That is a special case. The special case: when AB is a diameter, $AOB = 180^\circ$, so the inscribed angle $= 180^\circ/2 =$ _____ $^\circ$.

Angles in a semicircle:

If AB is a diameter, then any inscribed angle $APB =$ _____ $^\circ$.

Finding unknown angles:

In a circle, central angle $AOB = 110^\circ$. Find:

a) Inscribed angle APB (same arc) = _____ $^\circ$

b) Inscribed angle AQB subtending the **major** arc AB .

(Hint: major arc central angle $= 360^\circ - 110^\circ =$ _____ $^\circ$) $AQB =$ _____ $^\circ/2 =$ _____ $^\circ$

Retrieval Check — Close your notes

1. State the inscribed angle theorem in your own words.

2. A central angle is 144° . What is the inscribed angle subtending the same arc?

3. An inscribed angle is 23° . What is the central angle subtending the same arc?

4. Why do all inscribed angles subtending the same arc equal each other?

5. (Stretch) Arc CD has central angle 130° . Find inscribed angle CPD.

Self-check: ☐ I could teach it ☐ Getting there ☐ Not yet

Section 4 — Tangent-Radius Property

Learning target: I can apply the tangent-radius perpendicularity property and use it with the Pythagorean theorem to solve problems.

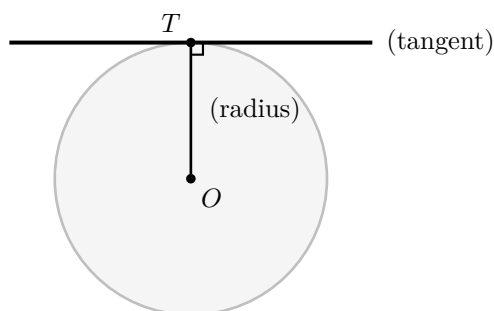
VOCABULARY

Term	Definition in your own words	One example
Tangent		
Point of tangency		
Perpendicular		
Tangent from external point		

GUIDED NOTES

Tangent-Radius Property:

A tangent to a circle is _____ to the radius at the point of tangency.

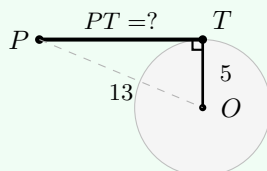


Angle $OTP =$ _____ $^\circ$

This means: if you know a line is tangent, the angle at the point of tangency is automatically _____.

Worked Example

A tangent PT touches a circle with centre O and radius 5 cm. Point P is 13 cm from the centre. Find PT .



Right angle at _____ (tangent $\perp$ radius) Hypotenuse = $PO =$ _____

$$PT^2 = PO^2 - OT^2 = 13^2 - 5^2 = \underline{\hspace{2cm}} - \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$$

$$PT = \underline{\hspace{2cm}} \text{ cm}$$

Explain: Why is PO the hypotenuse?

Equal tangents from an external point:

From any external point P, two tangents can be drawn to a circle. The two tangent segments are _____ in length.

$$PT_1 = \underline{\hspace{2cm}}$$

Alberta context: A circular fountain in a Calgary park has radius 3 m. A path runs tangent to the fountain, and the nearest point on the path is 5 m from the fountain's centre. How long is the section of path from where it first touches the tangent line to where it last could touch? (*Hint: find the tangent length, then consider symmetry.*)

Retrieval Check — Close your notes

1. State the tangent-radius property.

2. A tangent is drawn to a circle of radius 6 cm. The external point is 10 cm from the centre. Find the tangent length.

3. From an external point P, two tangents are drawn. One segment is 8 cm. What is the length of the other?

4. Why does the tangent-radius property create a right triangle? How is this useful?

5. (*Stretch*) Circle with centre O, radius 7 cm. Tangent at T, external point P where $OP = 25$ cm. Find PT. Then find angle OPT to the nearest degree.

Self-check: ☐ I could teach it ☐ Getting there ☐ Not yet

Unit Synthesis

Fill this in from memory. Notes closed.

PART A — KEY IDEAS

1. State the two chord properties from Section 2.

2. State the inscribed angle theorem. What is the special case when the chord is a diameter?

3. State the tangent-radius property. How does it allow you to use the Pythagorean theorem?

PART B — PRACTISE FROM MEMORY

In a circle with centre O and radius 10 cm: chord $AB = 16$ cm, inscribed angle $APB = 38^\circ$.

Find: a) the distance from O to chord AB. b) the central angle AOB.

Show working:

PART C — GO BACK TO YOUR PRE-KNOWLEDGE CHECK

Read what you wrote before this unit started.

1. How many circle parts did you label correctly? Which ones did you not know?

2. What is an inscribed angle? How does it compare to your guess?
